Garage opener carrier PCB — Bill of Materials

Rev 1 · 26 fitted parts · generated 2026-10-16 from hardware/garage-opener-pcb/bom.csv · schematic: docs/Schematic_GarageOpener.pdf

REFS	QTY	VALUE	FOOTPRINT	LCSC
C1	1	100uF 16V	CP_Radial_D6.3mm_P2.50mm	–
C2-C4	3	100nF	C_0805_2012Metric	C49678
D1	1	SS34	D_SMA	C8678
D2	1	1N4148W	D_SOD-123	C81598
D3	1	PESD5V0S1BA	D_SOD-323	–
D4	1	RUN green	LED_D3.0mm	–
D5	1	DOOR amber	LED_D3.0mm	–
F1	1	1.1A hold	Fuse_1812_4532Metric	–
J1	1	5V IN	PhoenixContact_MC_1,5_2-G-3.81_1x02_P3.81mm_Horizontal	–
J2	1	OPENER COM/NO/NC	PhoenixContact_MSTBA_2,5_3-G-5,08_1x03_P5.08mm_Horizontal	–
J3	1	REED	PhoenixContact_MSTBA_2,5_2-G-5,08_1x02_P5.08mm_Horizontal	–
K1	1	SRD-05VDC-SL-C	Relay_SPDT_SANYOU_SRD_Series_Form_C	C35449
Q1	1	MMBT2222A	SOT-23	–
R1,R4	2	1k	R_0805_2012Metric	C17513
R2,R5	2	4k7	R_0805_2012Metric	C17673
R3,R9,R11	3	10k	R_0805_2012Metric	C17414
R6,R7	2	330	R_0805_2012Metric	C17630
U1	1	ESP32-DevKitC-V4	ESP32-DevKitC-V4_Socket	–
U2	1	DHT22	PinSocket_1x04_P2.54mm_Vertical	–

Not populated (DNP)

REFS	VALUE	FOOTPRINT	LCSC	NOTE
R8, R10	10k	R_0805_2012Metric	C17414	Board-ID straps to +3V3 — fit only on future revisions

Also needed (no schematic line)

QTY	ITEM	NOTE
1	ESP32 DevKitC V4 (ESP32-WROOM-32U)	Plugs into U1. Needs an external 2.4 GHz antenna on its U.FL connector.
2	Female header 1x19, 2.54 mm	U1 sockets (rows 25.4 mm apart)
1	Plug Phoenix MC 1,5/2-ST-3,81 (or 15EDGK clone)	Mates with J1 (5 V in)
1	Plug Phoenix MSTB 2,5/2-ST-5,08 (or 2EDGK clone)	Mates with J3 (reed)
1	Plug Phoenix MSTB 2,5/3-ST-5,08 (or 2EDGK clone)	Mates with J2 (relay COM/NO/NC)
1	U.FL (IPEX1) to SMA-female bulkhead pigtail, ~10 cm + 2.4 GHz SMA antenna	Antenna through the enclosure wall
8	M3 heat-set insert, M3×5×4 mm	Enclosure: 4 for the PCB, 4 for the lid
4	M3×8 screw	PCB to base (H1–H4)
4	M3×10 countersunk screw	Lid
2	4 mm wood screw + wall plug	Wall bracket
1	5.1–5.2 V DC power supply	Into J1. A plain 5.0 V supply leaves too little margin.

LCSC numbers were verified on lcsc.com when the BOM was created; "—" means no number has been chosen yet. Before ordering a PCB, read the checklist in hardware/garage-opener-pcb/README.md.